Research Highlights

The Epistemic Fluency Trap in AI-Augmented Security Operations: Generative Rationale Fluency, Contested Evidence Cues, and Analyst Verification

Submitted to *Journal of Management Information Systems* (Taylor & Francis)

Concise Highlights (Each Bullet ≤ 85 Characters Including Spaces)

- Formalizes the Epistemic Fluency Trap via Bayesian decision theory (DCEDM).
- Unpartitioned lexical, neural MLP, and subword LM scorers collapse under ITPI.
- Prompt-isolated PITSA guarantees zero recall loss across 2.24M real logs.
- Selective Two-Stage DECS cuts warning banners by 79.9% to 95.9% across streams.
- Exact cost-ratio crossovers (κ^*) and U-shaped habituation optima derived.

Extended Executive Summary of Contributions for *JMIS*

1. **Dual-Cue Epistemic Defeasibility Model (DCEDM, Theorem 1 & Corollary 1):** Integrates metacognitive processing fluency theory, human-AI delegation, Toulmin defeasible argumentation, and endogenous neuro-cognitive warning habituation into a normative Bayesian decision-theoretic governance model for SLA-constrained SOC alert triage.

2. **Prompt-Isolated Telemetry Discrepancy Scorer (PITSA) & Two-Stage DECS (Proposition 1, DP1 to DP3):** Enforces a mathematical air-gap (Π_{struct}, $\nabla_{\mathcal{A}_{\text{text}}} L(e) \equiv 0$) between kernel/flow structural pillars ($\mathcal{A}_{\text{struct}}$) and attacker-controllable strings ($\mathcal{A}_{\text{text}}$), extracting exact per-pillar forensic defeaters ($k^*(e)$) in 2.12 to 4.31 μs per event on a single CPU core.

3. **Multi-Corpus Zero-Leakage Empirical Validation ($N = 2,238,434$ Records, $671,531$ Held-Out Test Records):** Across `OTRF Security-Datasets`, `SALAD-SOC`, `LMD-2023`, and `CISA KEV`, five unpartitioned lexical, dense semantic neural MLP, and subword language model classifiers suffer statistically significant recall collapse under graded appended ($m \in \{1, 2, 4, 8\}$) and full-override ITPI (McNemar's $\chi^2 \geq 86.01$ on OTRF and $\chi^2 \geq 8,361.46$ on SALAD-SOC, $p < 0.0001$), whereas Partitioned PITSA maintains 0.00 pp degradation (AUROC = 0.9807 on OTRF, 0.9380 on SALAD-SOC, and 0.9963 on LMD-2023).

4. **Trace-Driven Governance Dominance Boundaries and Habituation Optimization:** Selective DECS (Π_4) reduces Compromise Miss Rates by 18.45 pp (OTRF) and 37.32 pp (SALAD-SOC) at $\lambda_F = 0.40$, reduces expected operational loss by 30.16% to 60.77% on SALAD-SOC ($\kappa^* = 1.68$), and achieves an interior U-shaped optimal gating threshold at $\tau^*_{\text{LLR}} = 1.0$.